

Lecture 1: Majorana fermions in condensed matter

Sunday, February 8, 2015 2:55 PM

Title: Anyonic Defects: A New Paradigm for Non-Abelian Statistics

Plan for lectures:

1. Majorana fermions in condensed matter physics
2. Fractionalized Majoranas and parafermionic algebra on FQH edges
3. Generalized discussion of defects. FQH bilayers. Ising defects

Background

In the preceding lectures, you saw how to describe *topologically ordered states of matter*. These are $T=0$ phases of interacting particles (in most of these lectures we have focused on two spatial dimensions, although higher dimensional examples are also possible), which are gapped in their bulk. Nevertheless, there is a non-local (topological) order in the ground state wavefunction; the system cannot be adiabatically connected to a different, non-topologically ordered phase without crossing a phase transition.

Defining properties of topological order:

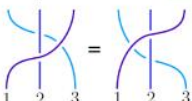
- Gap in the bulk, not adiabatically continuable to a "trivial" (site-factorizable) phase
- Topological degeneracy on high-genus surfaces: the g.s. wavefunction "feels" the topology of the manifold. This degeneracy is topological, in the sense that it cannot be lifted by any local term: the degenerate states are distinct by a "non-local" (string) operator. Only way to lift the degeneracy is through a phase transition, in which the bulk gap closes.
- The phase supports fractionalized "particle-like" excitations which carry fractional statistics. These are often associated also with fractionalized quantum numbers (as the $e/3$ charges in the Laughlin FQ state). The statistics of the excitations can be non-Abelian (more on this later). The presence of these excitations is directly related to the degeneracy of the g.s. on high-genus surfaces.
- In some (but not all) cases: the topological order may manifest itself in the presence of gapless edge states in a system with open boundary conditions. These gapless edge states may be described by a CFT. E.g. quantum Hall phases discussed in other lecture. NOT in phases such as Kitaev's toric code (the Z_2 gauge theory).

Non-Abelian statistics arises if the topological degeneracy of the ground state *increases exponentially* when excitations are introduced. In this situation, exchanging two excitations ("anyons") corresponds not to a phase factor, but a *matrix* acting on the ground state Hilbert space.

Non-Abelian Statistics: Braiding

$|\psi_i\rangle \rightarrow \sum_j U_{ij} |\psi_j\rangle$



Braid group:  = 

Majorana Fermions: $e^{(\pi/4)\gamma_1\gamma_2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}$

2D vortices: Ivanov, Read & Green, ...
1D wire network: Alicea et.al (2010)

There is clearly a connection between the topological degeneracy that appears on a non-zero genus surface and the one that appears when non-Abelian excitations are introduced. In these lectures, I will explore this connection. I will show that introducing defects into these systems mimics the effect of a high-genus surface. As a result, in the presence of these defects, the ground state degeneracy increases. Moreover, one can define a "braiding" operation on defects; they give rise to a new kind of non-Abelian statistics, distinct from that of anyons. Defects in Abelian systems behave as non-Abelian

objects; introducing defects in a non-Abelian system enriches its properties, providing additional "quantum gates" that can be implemented in a topologically protected manner.

Majorana zero modes in condensed matter physics

Majorana zero modes are at the heart of the simplest kind of non-Abelian statistics - that of Ising anyons - and also the simplest example of a non-Abelian defect. They are also the kind which is probably closest to experimental realization. I will spend most of this lecture describing them.

Majorana zero modes are closely tied with superconductivity. They can appear at cores of vortices in p-wave superconductors, and at edges of superconducting wires. They also appear as quasi-particles of the Moore-Read $\nu = 5/2$ state; this state turns out to be just a paired (superconducting) state of composite fermions. Let me start by briefly remind you some basic facts about the spectrum of a superconductor.

Bogoliubov - de Gennes (BdG) Formalism

For concreteness, consider a model of spinless fermions on a d -dimensional lattice.

$$\{\psi_i, \psi_j\} = 0, \quad \{\psi_i^\dagger, \psi_j\} = \delta_{ij}$$

If there are attractive interactions of the form (schematically) $-\psi_i^\dagger \psi_j^\dagger \psi_k \psi_l$, the system may become unstable to a "condensate" where the fermion pair operator $O_{ij} = \psi_i \psi_j$ becomes long-range ordered: $\langle O_{ij} O_{kl}^\dagger \rangle \neq 0$ when k, l are far from i, j .

Mean field approximation (BCS)

$$\langle O_{ij} \rangle = \langle \psi_i \psi_j \rangle \neq 0$$

$$\text{mean-field decoupling} \quad \langle \psi_i^\dagger \psi_j^\dagger \psi_k \psi_l \rangle \rightarrow \langle \psi_i^\dagger \psi_j^\dagger \rangle \psi_k \psi_l + \langle \psi_k \psi_l \rangle \psi_i^\dagger \psi_j^\dagger$$

$\Rightarrow$ "Anomalous" terms that do not conserve the number appear.

BCS Mean-field Hamiltonian:

$$H = \underbrace{\sum_{i,j} -t_{ij} \psi_i^\dagger \psi_j}_{\text{hopping}} - \underbrace{\mu \sum_i \psi_i^\dagger \psi_i}_{\text{chemical potential}} + \underbrace{\sum_{i,j} \Delta_{ij} c_i^\dagger c_j^\dagger + \text{h.c.}}_{\text{pairing potential}}$$

$$= (\psi_1^\dagger \dots \psi_N^\dagger \quad \psi_1 \dots \psi_N) \begin{pmatrix} h_{ij} & \Delta_{ij} \\ \Delta_{ij}^* & h_{ij} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix} \Bigg\} \Psi$$

$$= \underbrace{(\psi_1^\dagger \dots \psi_N^\dagger \quad \psi_1 \dots \psi_N)}_{\Psi^+ \text{ Nambu Spinor}} \begin{pmatrix} h_{ij} & \Delta_{ij} \\ (\Delta^\dagger)_{ij} & -h_{ij}^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \\ \psi_1^\dagger \\ \vdots \\ \psi_N^\dagger \end{pmatrix} \} \Psi$$

(Here $h_{ij} = \frac{1}{2}(-t_{ij} - \mu \delta_{ij})$ $\mathcal{H}: 2N \times 2N$)

Quadratic in Ψ 's $\rightarrow$ can be solved!

Bogoliubov transformation:

$$\begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \\ \psi_1^\dagger \\ \vdots \\ \psi_N^\dagger \end{pmatrix} \xrightarrow{\text{Unitary } U} \begin{pmatrix} f_1 \\ \vdots \\ f_N \\ f_1^\dagger \\ \vdots \\ f_N^\dagger \end{pmatrix} \} F$$

Check:

 $\{f_i^\dagger, f_j\} = \delta_{ij}$

$$\mathcal{H} = \Psi^\dagger \mathcal{H} \Psi = F^\dagger \underbrace{(U^\dagger \mathcal{H} U)}_{\text{Choose } U \text{ to diagonalize } \mathcal{H}} F$$

Choose U to diagonalize $\mathcal{H}$:

$$U^\dagger \mathcal{H} U = \begin{pmatrix} E_1 & & \\ & \ddots & \\ & & E_{2N} \end{pmatrix}$$

Symmetry of the BdG Hamiltonian:

$$C^{-1} \mathcal{H} C = -\mathcal{H}, \text{ where } C = \begin{pmatrix} 0 & \mathbb{1}_{N \times N} \\ \mathbb{1}_{N \times N} & 0 \end{pmatrix} \underbrace{K}_{\text{Complex conjugation}}$$

"Particle-hole" symmetry

(You can prove this by direct substitution

$$\begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} K \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^* \end{pmatrix} \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} K = K \begin{pmatrix} -h^* & \Delta^\dagger \\ \Delta & h \end{pmatrix} K$$

$$= \begin{pmatrix} -h & \Delta^T \\ \Delta^* & h^* \end{pmatrix}$$

You can check that $\Delta^T = -\Delta$

$$\Delta^* = (\Delta^T)^\dagger = -\Delta^\dagger$$

since if Δ is symmetric, $\Delta_{ij} c_i^\dagger c_j^\dagger = 0$

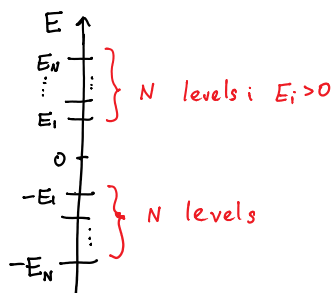
$$\Rightarrow \dots = - \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^* \end{pmatrix}.$$

$$\langle \psi | H | \psi \rangle = \langle \psi | H | \psi \rangle$$

Note also that $\Psi = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_n \\ \psi_1^\dagger \\ \vdots \\ \psi_n^\dagger \end{pmatrix} = C \begin{pmatrix} \psi_1^\dagger \\ \vdots \\ \psi_n^\dagger \\ \psi_1 \\ \vdots \\ \psi_n \end{pmatrix} \equiv C \Psi^*$

This implies that the levels of H come in pairs of opposite sign: if $H u = E u$, then

$$H(Cu) = C(\underbrace{C^{-1} H C}_{-H}) u = -E C u$$



The levels at E_i and $-E_i$ are not independent; they are "two halves of the same state".

After the transformation: $H = F^\dagger \begin{pmatrix} E_1 & & & \\ & \ddots & & \\ & & E_N & \\ & & & -E_1 & & \\ & & & & \ddots & \\ & & & & & -E_N \end{pmatrix} F$

$$= \sum_{i=1}^N E_i (f_i^\dagger f_i - f_i f_i^\dagger)$$

$$= \sum_{i=1}^N E_i (2 f_i^\dagger f_i - 1)$$

Many-body ground state: $f_i^\dagger f_i |0\rangle = 0 \quad \forall i=1 \dots N$
(since $E_i > 0$)

Excited states: $\prod_{j=1}^n f_{\alpha_j}^\dagger \dots f_{\alpha_n}^\dagger |0\rangle$

Energy (rel. to ground state)
 $2(E_{\alpha_1} + \dots + E_{\alpha_n})$

Example: p+ip superconductor in d=2

Take nearest-neighbor hopping: $\sum_{i,j} -t_{ij} c_i^\dagger c_j$

$$= -t \sum_{\langle i,j \rangle} c_i^\dagger c_j + \text{h.c.}$$

n.n. hopping

In momentum space

$$\psi_{\vec{k}} = \frac{1}{\sqrt{N}} \sum_i e^{i\vec{k} \cdot \vec{r}_i} \psi_i$$

$$h = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^{\dagger} \left(-2t (\cos k_x + \cos k_y) - \mu \right) \Psi_{\mathbf{k}}$$

$$\xrightarrow{|\mu + 4t| \ll t} \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^{\dagger} \left(\frac{\vec{k}^2}{2m} - \tilde{\mu} \right) \Psi_{\mathbf{k}} + \text{const.}$$

$$\frac{1}{m} = 2t \quad \tilde{\mu} = \mu - 4t$$

(From now on replace $\tilde{\mu} \rightarrow \mu$)

$$\text{Nearest-neighbor pairing: } -i\Delta \sum_i (c_i^{\dagger} c_{i+\hat{x}}^{\dagger} + i c_i^{\dagger} c_{i+\hat{y}}^{\dagger}) + \text{h.c.}$$

$$= \Delta \sum_{\mathbf{k}} \underbrace{(\sin k_x + i \sin k_y)}_{\approx k_x + i k_y \text{ for } |\vec{k}| \ll 2\pi} c_{\mathbf{k}}^{\dagger} c_{-\mathbf{k}}^{\dagger} + \text{h.c.}$$

$$\left(\sum_i c_i^{\dagger} c_{i+1}^{\dagger} = \frac{1}{N} \sum_{\mathbf{k}, \mathbf{k}'} \sum_i e^{-i\mathbf{k} \cdot \mathbf{x}_i - i\mathbf{k}' \cdot (\mathbf{x}_i + \hat{x})} c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}'}^{\dagger} \right)$$

$$= \sum_{\mathbf{k}} e^{+i\mathbf{k}} c_{\mathbf{k}}^{\dagger} c_{-\mathbf{k}}^{\dagger} = \frac{1}{2} \sum_{\mathbf{k}} (e^{i\mathbf{k}} - e^{-i\mathbf{k}}) c_{\mathbf{k}}^{\dagger} c_{-\mathbf{k}}^{\dagger} = i \sum_{\mathbf{k}} \sin \mathbf{k} c_{\mathbf{k}}^{\dagger} c_{-\mathbf{k}}^{\dagger}$$

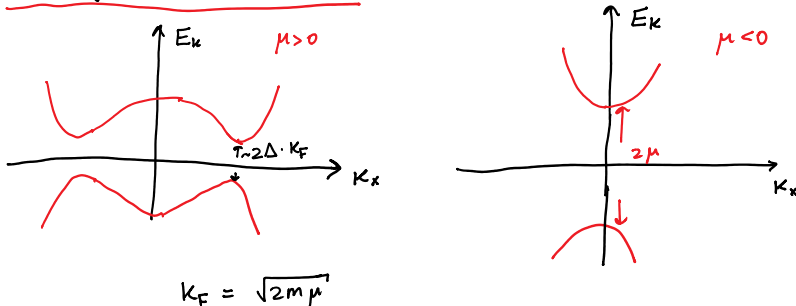
The Hamiltonian in momentum space has the form

$$H = \frac{1}{2} \sum_{\mathbf{k}} \begin{pmatrix} \Psi_{\mathbf{k}} & \Psi_{-\mathbf{k}}^{\dagger} \end{pmatrix} h_{\mathbf{k}} \begin{pmatrix} \Psi_{\mathbf{k}} \\ \Psi_{-\mathbf{k}}^{\dagger} \end{pmatrix}$$

$$h_{\mathbf{k}} = \begin{pmatrix} \frac{k^2}{2m} - \mu & \Delta(k_x + i k_y) \\ \Delta(k_x - i k_y) & -\left(\frac{k^2}{2m} - \mu\right) \end{pmatrix}$$

$$\text{Diagonalize: } E_{\mathbf{k}, \pm} = \pm \sqrt{\left(\frac{k^2}{2m} - \mu\right)^2 + \Delta^2 |k_x + i k_y|^2}$$

Distinguish two cases: $\mu < 0$ and $\mu > 0$



"weak pairing" $\longleftrightarrow$ "Strong pairing" ($\Leftrightarrow$ vacuum)

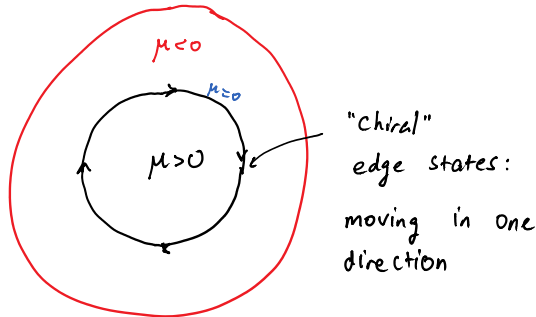
Gap closes on the way at $\vec{k} = 0$

These phases are topologically distinct: have a different Chern number.

We will consider the edge of the system.

Edge states consider a system with a disc geometry, where μ changes

Continuously:



$$x = r \cos \theta \quad dx = dr \cos \theta - r \sin \theta d\theta$$

$$y = r \sin \theta \quad dy = dr \sin \theta + r \cos \theta d\theta$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y} = -r \sin \theta \frac{\partial}{\partial x} + r \cos \theta \frac{\partial}{\partial y}$$

$$\begin{pmatrix} \partial_r \\ \frac{1}{r} \partial_\theta \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix}$$

$$\partial_x = \cos \theta \partial_r - \frac{1}{r} \sin \theta \partial_\theta$$

$$\partial_y = \sin \theta \partial_r + \frac{1}{r} \cos \theta \partial_\theta$$

$$\partial_x + i \partial_y = (\cos \theta + i \sin \theta) \partial_r + \frac{1}{r} (-\sin \theta + i \cos \theta) \partial_\theta$$

$$= e^{i\theta} \partial_r + \frac{i e^{i\theta}}{r} \partial_\theta$$

$$-\partial_x + i \partial_y = (-\cos \theta + i \sin \theta) \partial_r + \frac{1}{r} (\overbrace{+\sin \theta + i \cos \theta}^{i(\cos \theta - \sin \theta)}) \partial_\theta$$

$$= -e^{-i\theta} \partial_r + \frac{i e^{-i\theta}}{r} \partial_\theta$$

Assume very slow spatial variation, keep only constants
and $\partial_{x,y}$ (neglect ∇^2)

BdG equation:

$$\begin{pmatrix} -\mu(r) & \Delta e^{i\theta} \left(\partial_r + \frac{i}{r} \partial_\theta \right) \\ \Delta e^{-i\theta} \left(-\partial_r + \frac{i}{r} \partial_\theta \right) & \mu(r) \end{pmatrix} \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix} = E \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix}$$

$$\begin{pmatrix} -\mu(r) & \Delta e^{i\theta} \left(\partial_r + \frac{i}{r} \partial_\theta \right) \\ \Delta e^{-i\theta} \left(-\partial_r + \frac{i}{r} \partial_\theta \right) & +\mu(r) \end{pmatrix} \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix} = E \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix}$$

$$\left. \begin{aligned} u &= e^{i(m+1)\theta} \frac{f(r)}{\sqrt{r}} \\ v &= e^{im\theta} \frac{g(r)}{\sqrt{r}} \end{aligned} \right\} \text{ Ansatz} \quad \partial_r (r^{-1/2} f) = -\frac{1}{2} r^{-3/2} f + r^{-1/2} \partial_r f = r^{-1/2} \left(\partial_r - \frac{1}{2r} \right) f$$

$$\begin{pmatrix} -\mu(r) & \Delta \left(\partial_r - \frac{m+\frac{1}{2}}{r} \right) \\ \Delta \left(-\partial_r - \frac{m+\frac{1}{2}}{r} \right) & +\mu(r) \end{pmatrix} \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = E \begin{pmatrix} f(r) \\ g(r) \end{pmatrix}$$

assume $R \gg \frac{\Delta}{\mu} \Rightarrow$ replace $r \rightarrow R$

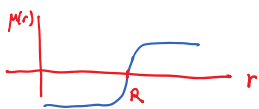
For simplicity take $\mu(r) = \begin{cases} +\mu_0 & r < R \\ -\mu_0 & r > R \end{cases}$
($\mu_0 > 0$)

We will first find solutions by setting $\Delta \frac{m+\frac{1}{2}}{R} \rightarrow 0$,
and then compute its effects perturbatively.

Zeroth order in $\frac{1}{R}$:

$$\begin{pmatrix} -\mu(r) \sigma^z + \Delta i \partial_r \sigma^y \\ \end{pmatrix} \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = E \begin{pmatrix} f(r) \\ g(r) \end{pmatrix}$$

Dirac equation with "kink" in mass term $\mu(r)$



Jackiw - Rebbi (70's): zero energy solution localized near $r = R$!

To find the zero energy solution, write

$$\begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} e^{-\alpha_\pm (r-R)} \quad \text{with } \alpha_\pm \text{ for } r > R, r < R \text{ respectively.}$$

$\Downarrow$ const. (ignore $\frac{m+\frac{1}{2}}{r}$ to 1st approx.)

$$r < R: \begin{pmatrix} -\mu_0 & -\alpha \Delta \\ \alpha \Delta & +\mu_0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

Solution: $\alpha_\pm \Delta = \pm \mu_0$

for $\Delta > 0$, pick $\alpha < 0$ solution so $f(r \ll R) \rightarrow 0$:

$$\begin{pmatrix} -\mu_0 & \mu_0 \\ -\mu_0 & +\mu_0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$a = +b$$

$$\text{solution: } \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = \begin{pmatrix} 1 \\ +1 \end{pmatrix} e^{+\frac{\mu_0}{\Delta}(r-R)} \equiv |\psi_0\rangle$$

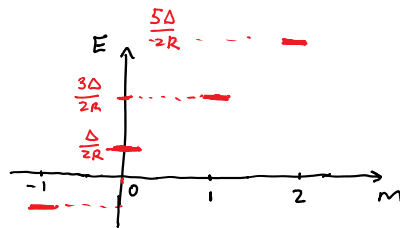
Similarly for $r > R$, pick $\alpha = +\frac{\mu_0}{R}$

$$\begin{pmatrix} +\mu_0 & -\mu_0 \\ \mu_0 & -\mu_0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$a = b$$

Now do 1st order perturbation theory in $\Delta \frac{m+\frac{1}{2}}{R}$:

$$E_m \approx \frac{\langle \psi_0 | \Delta \frac{m+\frac{1}{2}}{R} \sigma^x | \psi_0 \rangle}{\langle \psi_0 | \psi_0 \rangle} = \Delta \frac{m+\frac{1}{2}}{R} \quad m \in \mathbb{Z}$$



No $E=0$ solution!

However, is a vortex in the order parameter, all the levels shift by $\frac{1}{2} \frac{\Delta}{R}$ and there is a zero energy solution.

(A "vortex" means that the phase of the order parameter twists slowly in space, and changes by 2π when we go around a certain point, the vortex core).

in H_{delc} , change

$$\int dr \psi^\dagger(\vec{r}) \Delta (\partial_x + i\partial_y) \psi(\vec{r}) \rightarrow \int dr \psi^\dagger(r) e^{i\frac{\varphi(r)}{2}} \Delta (\partial_x + i\partial_y) e^{i\frac{\varphi(r)}{2}} \psi(r) + \text{h.c.}$$

$$\varphi(r, \theta) = -\theta \text{ "anti-vortex"}$$

Subtlety: when we write the pairing term $\int dr dr' \psi^\dagger(r) e^{i\frac{\varphi(r)}{2}} \Delta \partial_x \delta(r'-r) e^{i\frac{\varphi(r')}{2}} \psi(r')$ $\Delta(r, r')$ has to be anti-symmetric!

$$\varphi(r, \theta) = -\theta \quad \text{"anti-vortex"}$$

$\Delta(r, r')$ has to be anti-symmetric!

Inserting this into the BdG equation:

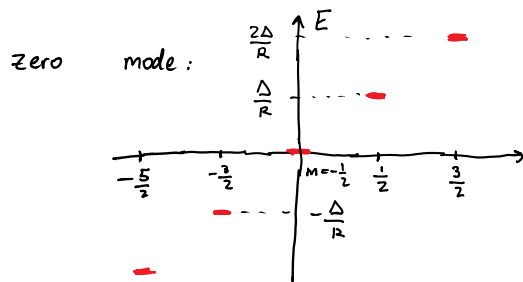
$$\begin{pmatrix} \mu(r) & e^{i\frac{\theta}{2}} \left(\partial_r + \frac{i}{r} \partial_\theta \right) e^{-i\frac{\theta}{2}} \\ \Delta e^{-i\frac{\theta}{2}} \left(-\partial_r + \frac{i}{r} \partial_\theta \right) e^{i\frac{\theta}{2}} & -\mu(r) \end{pmatrix} \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix} = E \begin{pmatrix} u(\vec{r}) \\ v(\vec{r}) \end{pmatrix}$$

To get rid of the phase factors, write $u = e^{-i\frac{\theta}{2}} \tilde{u}$ $v = e^{i\frac{\theta}{2}} \tilde{v}$. Then the Hamiltonian looks just as before, but the boundary conditions changed!

$$\tilde{u}(r, \theta + 2\pi) = -\tilde{u}(r, \theta) \quad (\text{and same for } \tilde{v})$$

Spectrum

$$E_n = \Delta \frac{m + \frac{1}{2}}{R}, \quad m \in \mathbb{Z} + \frac{1}{2}$$



Solutions to BdG have to come in pairs

$\Rightarrow$ Another zero mode at the vortex core!

Notes:

1. If we solved the model on the lattice, the number of solutions has to be even, since the dimension of the Hamiltonian is even. Therefore, there cannot be an isolated zero mode.
2. The edge modes are chiral, i.e., they move in one direction (similar to quantum Hall edges).
3. The $E = 0$ state (the zero mode) is special: the corresponding f operator is Hermitian, i.e. it is equal to its own conjugate:

Recall that the eigenmodes f_i of $\mathcal{H}$ are

$$f_i = u^\dagger \Psi, \quad f_i^\dagger = \Psi^\dagger u$$

where $u = \begin{pmatrix} 1 \\ z^N \\ 1 \end{pmatrix}$ is an eigenstate of $\mathcal{H}$.

A spatially isolated eigenstate of $\mathcal{H}$, $f_i = \tau$, satisfies

$\begin{pmatrix} 2N \\ 1 \end{pmatrix}$ is an eigenstate of τ_z .

A spatially isolated eigenstate of H , $\tau_z = \tau$, satisfies

$$u = C u = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} u^*$$

$$\Downarrow$$

$$\gamma = u^\dagger \Psi = u^\dagger \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_n \\ \psi_1^* \\ \vdots \\ \psi_n^* \end{pmatrix}$$

$$\gamma^\dagger = (\psi_1^\dagger - \psi_n^\dagger \quad \psi_1 - \psi_n) u = u^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_n \\ \psi_1^* \\ \vdots \\ \psi_n^* \end{pmatrix} = \gamma$$

$$\left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} u \right]^\dagger = (C u^*)^\dagger = (u^*)^\dagger = u^\dagger$$

γ is a so called "Majorana zero mode."

Note

The weak pairing phase of the p+ip SC is *topologically distinct* from the trivial phase. This can be seen from the fact that it has chiral edge states, or alternatively, from the fact that it has a zero mode at a vortex core ("defect"). Such an isolated mode at $E=0$ cannot disappear without a phase transition in which the bulk gap closes.

Extra note on Jackiw-Rebbi:

The $E=0$ solution can be found for arbitrary $\mu(r)$ as follows:

$$\left(-\mu(r) \sigma^z + \Delta i \partial_r \sigma^y \right) \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = 0$$

$\Downarrow$

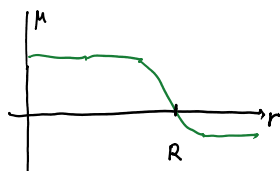
$$i \partial_r \begin{pmatrix} f \\ g \end{pmatrix} = i \sigma^x \frac{\mu(r)}{\Delta} \begin{pmatrix} f \\ g \end{pmatrix} = 0$$

$$\text{choose } \begin{pmatrix} f(r) \\ g(r) \end{pmatrix} = \begin{pmatrix} w(r) \\ w(r) \end{pmatrix}$$

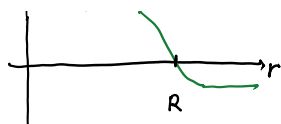
$$\partial_r w = \frac{\mu(r)}{\Delta} w$$

$$w(r) = w(0) e^{+ \int_0^r \frac{\mu(r)}{\Delta} dr}$$

I.e. for



I.e. for



we get

